Cell Tuition
1-1 PRIVATE TUTORS

NUMBER

Multiplying Decimals

Name _____ Date _____ No calculator Version 1

Scan to practise this topic online

KEY WORDS decimal *(tau ā-ira)* · decimal place *(mati ā-ira)* · product *(otinga whakarea)* · tenths *(hautekau)*

A A decimal times a whole number ●○○

Remember: Multiply as if there were no decimal point, then put it back: 0.6×4 is $6 \times 4 = 24$ with 1 decimal place, so 2.4.

1. Work out.

a) 0.6×4 = _____ **b)** 0.3×7 = _____ **c)** 1.2×3 = _____ **d)** 0.05×6 = _____

e) 2.4×5 = _____ **f)** 0.25×4 = _____ **g)** 3.1×8 = _____ **h)** 0.09×9 = _____

i) 1.5×12 = _____ **j)** 0.07×20 = _____ **k)** 4.25×2 = _____ **l)** 0.125×8 = _____

B A decimal times a decimal ●●○

Watch out: Count the decimal places in both numbers. 0.3×0.2 is $3 \times 2 = 6$ with 2 decimal places, so 0.06. It is not 0.6.

2. Work out.

a) 0.3×0.2 = _____ **b)** 0.5×0.7 = _____ **c)** 0.4×0.4 = _____ **d)** 1.2×0.3 = _____

e) 0.6×0.05 = _____ **f)** 2.5×0.4 = _____ **g)** 0.9×0.9 = _____ **h)** 1.1×1.1 = _____

i) 0.08×0.5 = _____ **j)** 3.2×0.2 = _____ **k)** 0.15×0.6 = _____ **l)** 1.5×2.5 = _____

m) 0.01×0.7 = _____ **n)** 4.5×0.02 = _____ **o)** 0.12×0.12 = _____ **p)** 6.4×0.5 = _____

C Fractions to decimals ●●○

Remember: A fraction is a division: $\frac{3}{4} = 3 \div 4 = 0.75$. These decimals stop, so they are called terminating decimals.

3. Write each fraction as a decimal.

a) $\frac{1}{2}$ **b)** $\frac{1}{4}$ **c)** $\frac{3}{5}$ **d)** $\frac{7}{10}$

e) $\frac{3}{4}$ **f)** $\frac{1}{8}$ **g)** $\frac{9}{20}$ **h)** $\frac{3}{25}$

i) $\frac{5}{8}$ **j)** $\frac{7}{4}$ **k)** $\frac{11}{50}$ **l)** $\frac{3}{40}$

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D Repeating decimals ●●●

> **Remember:** Some decimals never stop: $\frac{1}{3} = 0.3333\ldots$ A dot shows the digit that repeats: $0.\dot{3}$. Two dots show the first and last digits of a group that repeats.

4. Write each fraction as a decimal. Use dots to show the digits that repeat.

a) $\frac{1}{3}$ **b)** $\frac{2}{9}$ **c)** $\frac{1}{6}$ **d)** $\frac{5}{9}$

e) $\frac{2}{3}$ **f)** $\frac{5}{6}$ **g)** $\frac{3}{11}$ **h)** $\frac{1}{12}$

5. Write each fraction as a decimal, and say whether the decimal is terminating or repeating.

a) $\frac{3}{8}$ **b)** $\frac{1}{9}$ **c)** $\frac{7}{20}$ **d)** $\frac{4}{3}$

e) $\frac{8}{9}$ **f)** $\frac{3}{16}$ **g)** $\frac{7}{6}$ **h)** $\frac{2}{11}$

E Problems ●●●

6. Ribbon costs \$2.40 for each metre. How much does 3.5 m of ribbon cost?

7. A rectangle is 0.8 m long and 0.6 m wide. Find its area.

8. A bottle holds 1.25 litres of juice. How much juice is there in 6 bottles?

9. A snail moves 4.5 cm each second. How far does it move in 0.2 of a second?

★ Extension

10. You are told that $23 \times 14 = 322$. Use this to write down the answer to each calculation.

a) 2.3×14 **b)** 2.3×1.4 **c)** 0.23×14 **d)** 2.3×0.14

11. Which is larger, 0.6×0.6 or 0.6? Explain why.

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Multiplying Decimals

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS decimal *(tau ā-ira)* · decimal place *(mati ā-ira)* · product *(otinga whakarea)* · tenths *(hautekau)*

A A decimal times a whole number ●○○

> **Remember:** Multiply as if there were no decimal point, then put it back: 0.6×4 is $6 \times 4 = 24$ with 1 decimal place, so 2.4.

1. Work out.

a) 0.7×3 = _____ **b)** 0.4×8 = _____ **c)** 1.3×4 = _____ **d)** 0.06×5 = _____

e) 3.2×5 = _____ **f)** 0.75×4 = _____ **g)** 2.1×9 = _____ **h)** 0.08×7 = _____

i) 2.5×12 = _____ **j)** 0.03×30 = _____ **k)** 3.75×2 = _____ **l)** 0.375×8 = _____

B A decimal times a decimal ●●○

> **Watch out:** Count the decimal places in both numbers. 0.3×0.2 is $3 \times 2 = 6$ with 2 decimal places, so 0.06. It is not 0.6.

2. Work out.

a) 0.4×0.2 = _____ **b)** 0.6×0.7 = _____ **c)** 0.3×0.3 = _____ **d)** 1.3×0.2 = _____

e) 0.8×0.05 = _____ **f)** 1.5×0.4 = _____ **g)** 0.8×0.8 = _____ **h)** 1.2×1.2 = _____

i) 0.06×0.5 = _____ **j)** 4.1×0.2 = _____ **k)** 0.25×0.6 = _____ **l)** 3.5×1.5 = _____

m) 0.01×0.9 = _____ **n)** 3.5×0.02 = _____ **o)** 0.11×0.11 = _____ **p)** 8.6×0.5 = _____

C Fractions to decimals ●●○

> **Remember:** A fraction is a division: $\frac{3}{4} = 3 \div 4 = 0.75$. These decimals stop, so they are called terminating decimals.

3. Write each fraction as a decimal.

a) $\frac{1}{5}$ **b)** $\frac{3}{4}$ **c)** $\frac{2}{5}$ **d)** $\frac{9}{10}$

e) $\frac{1}{4}$ **f)** $\frac{3}{8}$ **g)** $\frac{7}{20}$ **h)** $\frac{6}{25}$

i) $\frac{7}{8}$ **j)** $\frac{9}{4}$ **k)** $\frac{13}{50}$ **l)** $\frac{7}{40}$

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D Repeating decimals ●●●

> **Remember:** Some decimals never stop: $\frac{1}{3} = 0.3333\ldots$ A dot shows the digit that repeats: $0.\dot{3}$. Two dots show the first and last digits of a group that repeats.

4. Write each fraction as a decimal. Use dots to show the digits that repeat.

a) $\frac{2}{3}$ **b)** $\frac{4}{9}$ **c)** $\frac{5}{6}$ **d)** $\frac{7}{9}$

e) $\frac{1}{3}$ **f)** $\frac{1}{6}$ **g)** $\frac{4}{11}$ **h)** $\frac{5}{12}$

5. Write each fraction as a decimal, and say whether the decimal is terminating or repeating.

a) $\frac{5}{8}$ **b)** $\frac{8}{9}$ **c)** $\frac{3}{20}$ **d)** $\frac{5}{3}$

e) $\frac{1}{9}$ **f)** $\frac{5}{16}$ **g)** $\frac{11}{6}$ **h)** $\frac{5}{11}$

E Problems ●●●

6. Ribbon costs \$3.20 for each metre. How much does 2.5 m of ribbon cost?

7. A rectangle is 0.9 m long and 0.4 m wide. Find its area.

8. A bottle holds 1.75 litres of juice. How much juice is there in 6 bottles?

9. A snail moves 6.5 cm each second. How far does it move in 0.4 of a second?

★ Extension

10. You are told that $26 \times 15 = 390$. Use this to write down the answer to each calculation.

a) 2.6×15 **b)** 2.6×1.5 **c)** 0.26×15 **d)** 2.6×0.15

11. Which is larger, 0.4×0.4 or 0.4? Explain why.

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